HALIC UNIVERSITY / FACULTY OF ENGINEERING / DEPARTMENT OF COMPUTER ENGINEERING
2025 - 2026 ACADEMIC YEAR / FALL SEMESTER / MTH1300 - NUMERICAL ANALYSIS
FINAL EXAM - 03/12/2025

Q3. Let $f(x) = x^2 e^x$.

a. Calculate approximations to $f'(2.3)$ using the central difference formula with $h = 0.01$ and $h = 0.001$ Carry eight decimal places. (15 pts)

b. Compare with the exact value of $f'(2.3)$. Which result is better? Explain briefly. (5 pts)

Answer:

91

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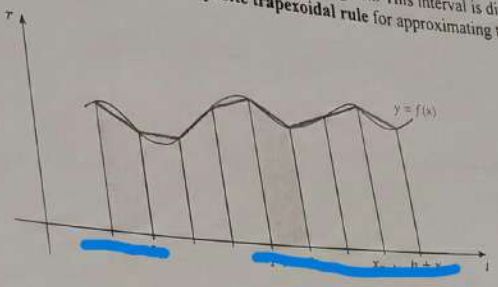
$$I = \int_0^{\pi} \sin^2 x dx$$

using the following methods:

a) Simpson's 1/3 method. Divide the whole interval into six subintervals.

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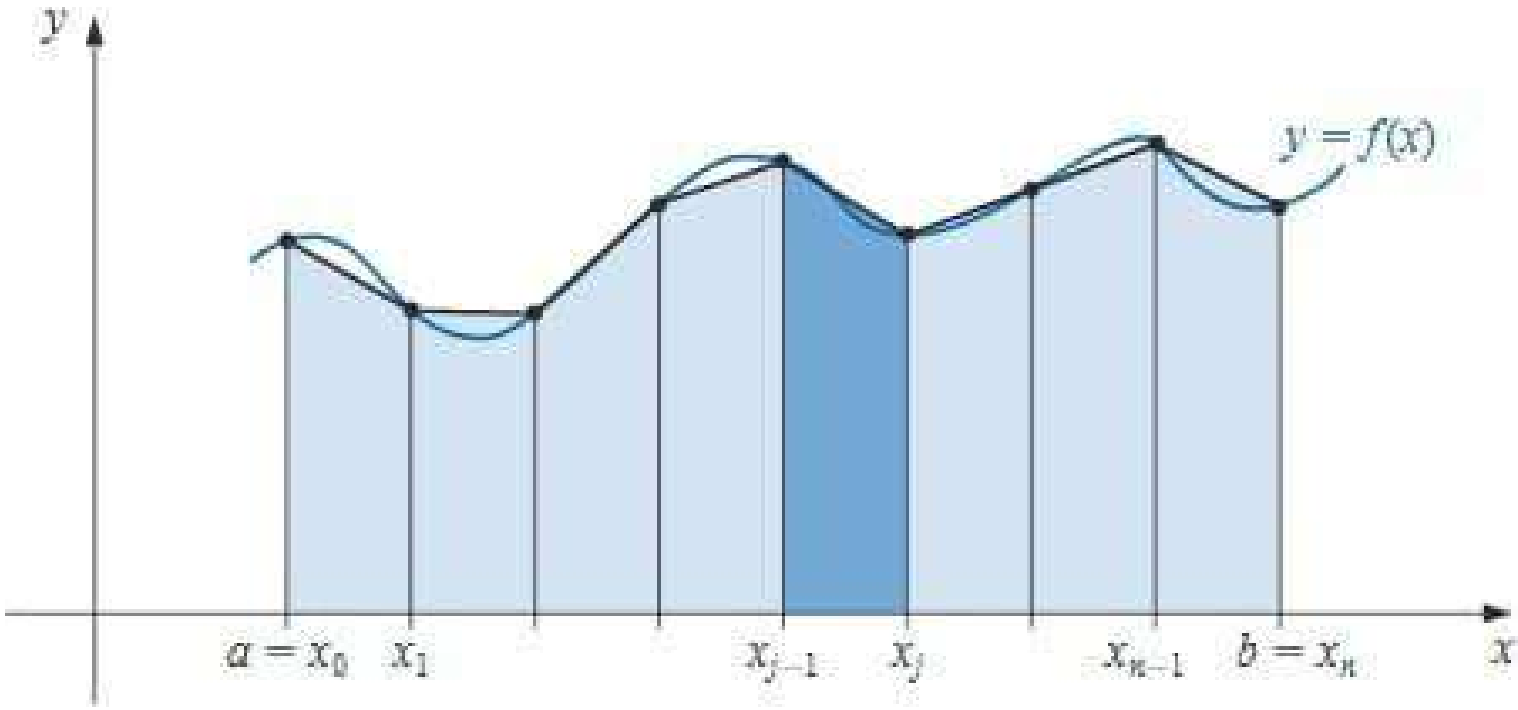
Q4. A function $f(x)$, defined and continuous on the interval $[a,b]$ is given. This interval is divided into n equal subintervals. Using this partition, **derive the composite trapezoidal rule** for approximating the definite integral $\int_a^b f(x)\,dx$.



Answer:

$F(x)$

$\int_a^b F(x)\,dx.$ (a, b)


y
y = f(x)
a = x0
x1
xj−1
xj
xn−1
b = xn
x

etimes taking a leap forward means leaving a few things behind!

$$P_3(x) = \frac{x^3 - 3x^2 + 2x}{0} - \frac{3x^3 - 3x^2 - 6x}{6} + \frac{8x^3 - 8x}{6}$$

$$= \frac{x^3 - 3x^2 + 2x - 3x^3 + 3x^2 + 6x + 8x^3 - 8x}{6}$$

$$= \frac{6x^3}{6} = x^3$$

Ex $f(x) = \sqrt{x}$ $\quad x = 4.5$

k	x_k	$f(x_k)$
0	4	2
1	5	2.23607
2	6	2.44949
3	7	2.64575
4	8	2.82843

a)

		a_0		a_1		a_2		a_3		a_4
4		2								
			$\frac{2.23607-2}{5-4}$							
5	2.23607			0.23607						
			$\frac{2.44949-2.23607}{6-5}$		$\frac{0.21342-0.23607}{6-4}$					
6	2.44949			0.21342		-0.011325				
			$\frac{2.64575-2.44949}{7-6}$		$\frac{0.19626-0.21342}{7-5}$		$\frac{-0.011325-0.000858}{7-4}$			
7	2.64575			0.19626		-0.000858		-0.000915		
			$\frac{2.82843-2.64575}{8-7}$		$\frac{0.18268-0.19626}{8-6}$		$\frac{-0.000679+0.000858}{8-5}$		$\frac{0.000597-0.000915}{8-4}$	
8	2.82843			0.18268		-0.000679		0.000597		-0.0000795

b) $P_1(x) = a_0 + a_1(x - x_0) \simeq 2 + 0.23607(x-4)$

$P_2(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1)$

$= 2 + 0.23607(x-4) - 0.011325(x-4)(x-5)$

$P_3(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$

$= 2 + 0.23607(x-4) - 0.011325(x-4)(x-5) + 0.000915(x-4)(x-5)(x-6)$

$P_4(x) = 2 + 0.23607(x-4) - 0.011325(x-4)(x-5) + 0.000915(x-4)(x-5)(x-6) - 0.000795(x-4)(x-5)(x-6)(x-7)$

c) $x = 4.5$

$P_1(4.5) = 2.11804$

$P_2(4.5) = 2.12085$

$P_3(4.5) = 2.12121$

$P_4(4.5) = 2.12128$

d) Exact $f(x) = \sqrt{x} \overset{x=4.5}{\rightsquigarrow} f(4.5) = 2.121320$

They are within the range

Ex] $f(x) = e^x$

a) $f'(2.3)$ $h = 0.1, 0.01, 0.001$

$f'(x) = e^x$

Exact → $f'(2.3) = 9.9742$

Using $h = 0.1$ $e^{2.2} = 9.0250$ $e^{2.4} = 11.0232$

x_{i-1} x_i x_{i+1}

2.2 2.3 2.4

$$FD = f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} = \frac{11.0232 - 9.97242}{2.4 - 2.3} = 10.4889$$

$$BD = f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = \frac{9.9742 - 9.0250}{2.3 - 2.2} = 9.49\%169$$

$$CD = f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{x_{i+1} - x_{i-1}} = \frac{11.0232 - 9.0250}{2.4 - 2.2} = 9.99\%81$$

$P_4(x) = 2 + 0.23607(x-4) - 0.011325(x-4)(x-5) + 0.000915(x-4)(x-5)(x-6) - 0.0000795(x-4)(x-5)(x-6)(x-7)$

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$$FD = f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} = \frac{11.0232 - 9.9742}{2.4 - 2.3} = 10.4899$$

$$BD = f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} = \frac{9.9742 - 9.0250}{2.3 - 2.2} = 9.49\%169$$

$$CD = f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{x_{i+1} - x_{i-1}} = \frac{11.0232 - 9.0250}{2.4 - 2.2} = 9.99\%81$$